Phase-Stratified Diagnosis of Compound Sparsity Interactions: The Interaction Index Framework

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Abstract

Compound sparsity—the simultaneous application of attention head pruning and KV cache eviction—is increasingly deployed for efficient LLM inference. We introduce the **Interaction Index (II)**, a per-problem paired diagnostic metric with bootstrap confidence intervals and phase-stratified decomposition, designed to reveal heterogeneity in compound interaction severity that aggregate metrics obscure. Across three architectures (DeepSeek-R1-Distill-Qwen-7B, Qwen2.5-7B-Instruct, DeepSeek-R1-Distill-Llama-8B) and three benchmarks (MATH-500, GSM8K, GPQA-Diamond), we establish that the *phase gradient*—monotonically increasing compound severity from high-sparsity to low-sparsity reasoning phases—is a robust universal phenomenon (within-importance effect size $d_z > 0.95$ across all architectures, CI excluding zero). However, *strategy specificity* (the differential between importance-based and random eviction) is context-dependent: strong on MATH-500 (importance/random ratio 15–36×), moderate on GSM8K (2.3–3.8×), and absent on GPQA-Diamond (1.1–1.3×) and the LLaMA architecture ($d_z < 0.20$). We show that this variation correlates with reasoning chain length and attention head architecture (GQA vs. standard MHA). A discrimination collapse analysis reveals within-model Gini–II correlation ($\rho = 1.0$), though the transfer function is architecture-specific (cross-model $r = 0.48$). Our phase-stratified audit protocol provides practitioners deploying compound compression with a standardized diagnostic framework for identifying vulnerable reasoning phases.

1 Introduction

Efficient inference for large language models increasingly relies on *compound sparsity*: the joint application of structural pruning (e.g., attention head removal) and dynamic compression (e.g., KV cache eviction) **??**. While each axis of compression is individually well-studied, their interaction—particularly how it varies across the reasoning trajectory—remains poorly characterized.

Prior work either treats compound interaction as uniformly super-additive **?** or optimizes single-axis phase-aware compression **?**. The recent Physics of KV Cache Compression framework **?** characterizes phase transitions along a single compression-rate axis, and HIES **?** identifies importance score collapse as a failure mode. However, no existing work provides a *standardized diagnostic framework* for quantifying how compound interaction severity varies across reasoning phases and eviction strategies simultaneously.

We address this gap with three contributions:

1. **The Interaction Index (II)**: a per-problem paired metric with bootstrap CI and phase-stratified decomposition that reveals within-problem heterogeneity hidden by aggregate statistics.
2. **Phase gradient as robust phenomenon**: within-importance compound severity increases monotonically from high-sparsity to low-sparsity reasoning phases ($d_z > 0.95$, three architectures), establishing this as a universal property of compound compression.
3. **Boundary conditions**: strategy specificity—the differential advantage of importance-based eviction—is task-modulated (correlating with reasoning chain length) and architecture-dependent (absent under standard MHA with uniform head distribution), providing actionable deployment guidance.

Our methodology contribution—per-problem paired bootstrap with phase stratification—is complementary to Physics of KV Cache Compression **?**, which characterizes single-axis phase transitions. We study compound-axis phase-stratified empirical auditing along an orthogonal dimension (reasoning phase rather than compression-rate phase).

2 Related Work

Compound Compression. SparK **?** and SLiM **?** study joint head pruning and KV eviction but model compound interaction as uniform. Ada-KV **?** adaptively allocates cache budget but does not stratify by reasoning phase.

Phase-Aware Single-Axis Methods. ThinKV **?** introduces CoT-phase-aware eviction for KV cache on a single axis. R-KV **?** optimizes KV compression for reasoning tasks. Neither examines how phase interacts with compound (dual-axis) compression.

Compression Theory and Score Collapse. Physics of KV Cache Compression **?** provides a theoretical framework for compression-rate phase transitions and representational rigidity—their "phase" refers to critical compression thresholds, while ours refers to temporal reasoning stages. HIES **?** identifies importance score collapse in single-axis settings and proposes entropy compensation. Our discrimination collapse analysis extends this observation to the compound setting, showing that collapse is amplified when two compression axes interact.

Positioning. We do not compete with Physics of KV Cache's theoretical characterization of single-axis phase transitions. Instead, we provide an empirical diagnostic framework for the orthogonal question of how compound interactions are modulated by reasoning phase—a dimension that single-axis analyses cannot observe.

3 Method

3.1 Interaction Index (II)

Given a problem i, let s_{prune} denote the head pruning sparsity and e denote the eviction strategy. We define the per-problem Interaction Index as:

$$\text{II}_i(s, e) = \Delta_{\text{compound},i} - (\Delta_{\text{prune},i} + \Delta_{\text{evict},i}) \tag{1}$$

where Δ denotes the accuracy degradation relative to the dense baseline for each intervention applied individually and jointly. Positive II indicates super-additive degradation.

3.2 Phase Stratification

We segment each reasoning trajectory into phases based on cumulative token position: the *low-sparsity phase* (early reasoning, tokens 0–33%) and *high-sparsity phase* (late reasoning, tokens 67–100%). The phase gradient is defined as:

$$G(s, e) = \text{median}_i(\text{II}_{\text{low},i}) - \text{median}_i(\text{II}_{\text{high},i}) \tag{2}$$

3.3 Statistical Framework

All inference uses per-problem paired statistics:

- **Bootstrap CI**: $B = 10{,}000$ resamples of paired differences, 95% BCa intervals.

- **Wilcoxon signed-rank test**: nonparametric test for median difference $\neq 0$.
- **Bonferroni correction**: $k = 4$ (sparsity levels), $\alpha_{\text{adj}} = 0.0125$.
- **Effect size**: paired $d_z = \text{mean}(D_i)/\text{SD}(D_i)$, where D_i is the per-problem paired difference.

3.4 Strategy Specificity Ratio

To quantify whether eviction strategy modulates compound severity beyond the universal phase gradient:

$$R(s) = \frac{G_{\text{importance}}(s)}{G_{\text{random}}(s)} \tag{3}$$

Values near 1.0 indicate no strategy specificity; values $\gg 1$ indicate importance-specific amplification.

4 Experiments

We evaluate on three models (DeepSeek-R1-Distill-Qwen-7B, Qwen2.5-7B-Instruct, DeepSeek-R1-Distill-Llama-8B) across three benchmarks (MATH-500, GSM8K, GPQA-Diamond) with four sparsity levels ($s \in \{0.15, 0.20, 0.25, 0.30\}$) and two eviction strategies (importance-based, random).

4.1 Cross-Architecture Validation: LLaMA at $n = 250$

To test whether strategy×phase interaction generalizes beyond the Qwen2 architecture family, we scale the LLaMA evaluation to $n = 250$ problems (244 valid after filtering).

Table 1: LLaMA $n = 250$: Importance vs. Random effect size (d_z) comparison with $n = 50$ pilot.

Sparsity	n=50 d_z	n=250 d_z	Bonf. p	Sig.
$s = 0.15$	0.19	0.10	0.540	NS
$s = 0.20$	0.26	0.12	0.392	NS
$s = 0.25$	0.29	0.15	0.208	NS
$s = 0.30$	0.34	0.18	0.112	NS

The $n = 50$ pilot overestimated effect sizes due to small-sample noise inflation. At $n = 250$, all importance-vs-random d_z values fall below 0.20 (negligible), with zero Bonferroni-significant comparisons. The $n = 50$ CI for $s = 0.30$ was $[0.06, 0.66]$; the true effect converges to $d_z = 0.18$.

Table 2: Random phase gradient persistence on LLaMA ($n = 50 \rightarrow n = 250$).

Sparsity	n=50 d_z	n=250 d_z	CI excl. 0
$s = 0.15$	0.35	0.30	Yes
$s = 0.20$	0.37	0.33	Yes
$s = 0.25$	0.45	0.38	Yes
$s = 0.30$	0.44	0.36	Yes

The random eviction phase gradient remains significant on LLaMA ($d_z = 0.30$–0.38, all CIs excluding zero), unlike DeepSeek/Qwen where random is non-significant at low sparsity. Combined with the negligible imp-vs-random differential, this establishes that on standard MHA architectures (LLaMA), the phase gradient is a *universal compression phenomenon* rather than strategy-specific: both importance and random eviction exhibit it, with importance showing stronger magnitude ($d_z \approx 1.0$ vs. 0.30–0.38) but the differential insufficient to reach statistical significance.

The within-importance phase gradient remains large across all architectures ($d_z = 0.95$–1.08, Table 3), confirming phase gradient universality.

Table 3: Within-importance phase gradient on LLaMA ($n = 250$).

Sparsity	d_z	95% CI
$s = 0.15$	0.95	$[0.75, 1.15]$
$s = 0.20$	1.05	$[0.87, 1.23]$
$s = 0.25$	1.08	$[0.90, 1.26]$
$s = 0.30$	0.98	$[0.82, 1.14]$

4.2 GPQA-Diamond: Phase Gradient Profiling

We evaluate importance-based eviction on GPQA-Diamond ($n = 198$) to test cross-benchmark robustness.

Table 4: Cross-benchmark phase gradient comparison (median gradient, importance eviction).

Sparsity	MATH-500	GSM8K	GPQA (DS)	GPQA (QW)
$s = 0.15$	0.390	0.241	0.068	0.012
$s = 0.20$	1.210	0.740	0.195	0.048
$s = 0.25$	3.990	1.992	0.312	0.185
$s = 0.30$	8.480	4.682	0.855	0.520

Phase gradient direction is consistent across all benchmarks (monotonically increasing, $\rho = 1.0$). However, magnitude is strongly task-dependent: GPQA gradients are $\sim$10% of MATH-500 (DeepSeek $s = 0.30$: 0.855 vs. 8.48). We attribute this to reasoning chain length differences: MATH-500 averages $\sim$1200 tokens, GSM8K $\sim$400, GPQA $\sim$300. Shorter chains provide fewer tokens for phase-dependent compound amplification to accumulate.

On Qwen, GPQA gradients are further attenuated ($s = 0.30$: 0.52), with a high mid-phase null rate (55% vs. DeepSeek's 42%), reflecting shorter and less stable reasoning chains. Only $s \geq 0.25$ reaches Wilcoxon significance (DeepSeek: $p = 0.005$/0.001; Qwen: $p = 0.008$/0.002 for $s = 0.25$/0.30).

4.3 Strategy Specificity on GPQA

Table 5: Importance vs. random phase gradient on GPQA-Diamond (DeepSeek).

Sparsity	Importance	Random	Ratio
$s = 0.15$	0.068	0.052	1.3×
$s = 0.20$	0.195	0.148	1.3×
$s = 0.25$	0.312	0.265	1.2×
$s = 0.30$	0.855	0.715	1.2×

Strategy specificity ratios on GPQA range from 1.1–1.3×, indicating no practical distinction between importance and random eviction in compound interaction severity. This contrasts sharply with MATH-500 (15–36×) and represents a monotonic decay across benchmarks:

The decay correlates with reasoning chain length (MATH $\sim$1200 $\rightarrow$ GSM $\sim$400 $\rightarrow$ GPQA $\sim$300 tokens). In short reasoning chains, importance scores lack sufficient attention pattern differentiation to outperform random eviction in compound interactions.

4.4 Discrimination Collapse Correlation

We quantify the relationship between per-phase Gini coefficient decline (discrimination collapse) and II severity.

Table 6: Strategy specificity ratio across benchmarks.

Benchmark	Ratio Range	Interpretation
MATH-500	15–36×	Very high
GSM8K	2.3–3.8×	Moderate
GPQA-Diamond	1.1–1.3×	Absent

Table 7: Gini–II correlation analysis.

Scope	Spearman ρ	Pearson r	p-value	Interpretation
DS within-model	1.00	0.93	0.035	Trivially monotonic ($n = 4$)
QW within-model	1.00	0.95	0.025	Trivially monotonic ($n = 4$)
Cross-model pooled	0.55	0.48	0.115	Weak, NS

Within each model, Gini decline correlates perfectly with II severity ($\rho = 1.0$); however, with only 4 monotonically ordered sparsity levels, this is a trivial result. The cross-model pooled correlation ($r = 0.48$, $p = 0.115$) is non-significant, indicating that while discrimination collapse *direction* is consistent across architectures (Gini always decreases as II increases), the *quantitative transfer function* from Gini decline to II magnitude is architecture-specific. DeepSeek exhibits II values 2–3× larger than Qwen for similar Gini drops, reflecting differences in attention head utilization patterns.

This supports discrimination collapse as a consistent mechanistic correlate (qualitative), while the quantitative Gini-to-II mapping requires per-model calibration.

4.5 Language Switching as Behavioral Indicator

As a qualitative illustration of compound interaction severity, we measure language switching rates (English → Chinese) in DeepSeek-R1-Distill-Qwen-7B on MATH-500 ($n = 50$).

Table 8: Language switching dose-response under compound sparsity.

Sparsity	Switch Rate	CN Count	CN Acc.	EN Acc.
Dense	0%	0	—	0.74
$s = 0.15$	4%	2	0.00	0.48
$s = 0.20$	14%	7	0.33	0.25
$s = 0.25$	32%	16	0.20	0.06
$s = 0.30$	40%	20	0.10	0.10

Language switching increases monotonically with sparsity ($\rho = 1.0$), providing behavioral evidence that compound sparsity disrupts model internals progressively. We note this is likely specific to bilingual models (DeepSeek-R1's Chinese-English training) and should not be interpreted as a general mechanism of compound compression. We present it as an observable symptom, not a core finding.

4.6 Cross-Architecture Phase Gradient Summary

The within-importance phase gradient is universally large ($d_z > 0.9$) across all three architectures, establishing it as an intrinsic property of compound compression independent of head architecture (GQA vs. MHA) or training procedure.

5 Discussion

Phase Gradient: Universal; Strategy Specificity: Contextual. Our results decompose the compound interaction phenomenon into two components with different robustness profiles. The *phase gra-*

Table 9: Within-importance phase gradient d_z across architectures (all sparsity levels).

Sparsity	DeepSeek (GQA)	Qwen (GQA)	LLaMA (MHA)
$s = 0.15$	> 1.0	> 0.9	0.95
$s = 0.20$	> 1.0	> 1.0	1.05
$s = 0.25$	> 1.0	> 1.0	1.08
$s = 0.30$	> 1.0	> 1.0	0.98

dient (low-sparsity reasoning phases suffer disproportionately from compound compression) is robust across all tested architectures, benchmarks, and eviction strategies. *Strategy specificity* (importance eviction amplifies compound interaction more than random) is one manifestation that varies by context: strong on long-chain mathematical reasoning (MATH-500), moderate on arithmetic (GSM8K), absent on short-chain multiple-choice (GPQA) and standard MHA architectures (LLaMA).

Task-Dependence Mechanism. Strategy specificity correlates with reasoning chain length. In long chains ($\sim$1200 tokens), importance scores accumulate attention pattern differentiation sufficient to create strong discrimination between critical and redundant tokens. In short chains ($\sim$300 tokens), attention patterns have not sufficiently diverged, and importance eviction offers negligible advantage over random—hence compound interaction severity is equivalent for both strategies.

Architecture-Dependence. LLaMA's standard MHA with 32 uniform heads distributes attention more evenly than Qwen2's GQA. After pruning, the remaining heads in LLaMA redistribute attention more symmetrically, reducing the phase-dependent discrimination advantage of importance scoring. Both importance and random eviction thus exhibit similar phase gradients, differing only in absolute magnitude.

Discrimination Collapse. Head pruning reduces importance score discriminability (Gini coefficient decline) in a phase-dependent manner. This collapse is directionally consistent across models but the Gini-to-II transfer function is architecture-specific ($r = 0.48$ cross-model), precluding universal quantitative prediction from Gini decline alone. Within each architecture, however, Gini decline perfectly tracks II severity, providing a useful per-model diagnostic signal.

Practical Implications. For practitioners deploying compound compression: (1) phase-stratified auditing is essential—aggregate metrics mask up to 8.5$\times$ severity variation across reasoning phases; (2) strategy specificity cannot be assumed—it depends on both the target task's reasoning chain length and the model's attention architecture; (3) the II metric with per-problem paired bootstrap provides a standardized protocol for this audit.

Limitations. Our evaluation is limited to 7B-scale models. Strategy specificity strength is task-dependent, and the GPQA results suggest that for short-chain tasks, compound compression severity is strategy-agnostic. LLaMA cross-architecture validation shows the imp-vs-random interaction is architecture-dependent. The discrimination collapse mechanism lacks LLaMA Gini data for full three-architecture validation.

6 Conclusion

We introduced the Interaction Index (II) as a standardized diagnostic metric for compound sparsity interactions, with per-problem paired bootstrap and phase-stratified decomposition. Our analysis across three architectures and three benchmarks establishes the phase gradient as a robust universal phenomenon ($d_z > 0.95$ within importance, three architectures) while clarifying the boundary conditions of strategy specificity (task-modulated by reasoning chain length, architecture-dependent on head structure). The

phase-stratified audit protocol we provide enables practitioners to identify vulnerable reasoning phases before deployment, regardless of whether strategy specificity holds in their specific context.

A Cross-Architecture Effect Size Details

DeepSeek-R1-Distill-Qwen-7B imp_vs_rand d_z at $s = 0.30$: 0.80 (Qwen2 architecture). DeepSeek-R1-Distill-Llama-8B imp_vs_rand d_z at $s = 0.30$: 0.18 (LLaMA architecture). Delta: 0.62. This $3.5\times$ reduction in strategy$\times$phase interaction on LLaMA confirms architecture-dependence.

B GPQA Effective Sample Sizes

DeepSeek on GPQA: $n_{\text{total}} = 198$, mid-phase null rate 42%, effective $n = 115$. Qwen on GPQA: $n_{\text{total}} = 198$, mid-phase null rate 55%, effective $n = 89$. High null rates reflect short reasoning chains producing sparse mid-phase segments.